

INDU 211: Introduction to Production & Manufacturing Systems

Problem Solution Guide 03: Project Management & PERT / CPM Scheduling

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[!NOTE]

- **YouTube Video Tutorial:** [Watch Video 3: INDU 211 - PERT Chart](#)
- **Video ID:** [b2g1kZrEYtk](#) · **Duration:** 6:52
- **Target Exam Scope:** 🎯 **FINAL EXAM (Weeks 7–12, Chapters 14, 15, 8, 17)**
- **Curriculum Context:** Chapter 17 — Project Management

1. Project Description & Activity Precedence Table

An industrial engineering project proposal involves seven distinct activities (**A through G**). The durations (in weeks) and strict immediate predecessor prerequisites are summarized in the project table below:

Activity	Name / Description	Immediate Predecessors	Duration (Weeks)
A	Assess Customer Needs	None	2
B	Write & Submit Proposal	A	1
C	Obtain Client Approval	B	1
D	Preliminary Engineering Design	C	2
E	Prototype Fabrication	C	5
F	System Testing & Validation	D, E	5
G	Final Client Delivery	F	1

2. Activity-on-Node (AON) Network Construction

1. Start of Project:

- Activity **A** has no prerequisites. Draw node **A** with duration 2 weeks.

2. Sequential Precedence:

- Activity **B** (duration 1) requires **A** $\implies A \rightarrow B$.

- Activity **C** (duration 1) requires **B** $\implies B \rightarrow C$.
- 3. **Parallel Branching at Node C:**
 - Activities **D** and **E** both depend strictly upon **C** being completed.
 - Node **C** branches into two parallel trajectories:
 - Upper Branch: Node **D** (duration 2 weeks).
 - Lower Branch: Node **E** (duration 5 weeks).
- 4. **Network Re-Convergence at Node F:**
 - Activity **F** (duration 5) requires **both** D and E to be 100% finished before it can begin.
 - Both paths converge into Node **F**.
- 5. **Final Project Completion:**
 - Activity **G** (duration 1) depends upon **F** $\implies F \rightarrow G$.

3. Path Enumeration & Critical Path Identification

A path through the project network is any continuous sequence of activities from the project start (Node A) to the project conclusion (Node G):

Path Number	Activity Sequence	Week Durations	Total Cumulative Duration
Path 1	$A \rightarrow B \rightarrow C \rightarrow \mathbf{D} \rightarrow F \rightarrow G$	$2 + 1 + 1 + \mathbf{2} + 5 + 1$	12 Weeks
Path 2	$A \rightarrow B \rightarrow C \rightarrow \mathbf{E} \rightarrow F \rightarrow G$	$2 + 1 + 1 + \mathbf{5} + 5 + 1$	15 Weeks

The Critical Path Decision:

- The **Critical Path** is defined as the sequence of dependent activities with the **longest total duration** through the network. It dictates the minimum calendar time required to complete the entire project.
- **Critical Path:**

$$\mathbf{A \rightarrow B \rightarrow C \rightarrow E \rightarrow F \rightarrow G}$$

- **Minimum Project Completion Duration:**

$$\text{Total Duration} = \mathbf{15 \text{ Weeks}}$$

4. Total Slack & Delay Sensitivity Analysis

- **Why does the Critical Path route through E instead of D?**
 - Activity F cannot begin until **both** D and E are completed.
 - Even if Activity D finishes in 2 weeks, the project cannot advance to testing because Prototype Fabrication (E) requires 5 weeks.
- **Activity D Slack:**

$$\text{Total Slack for Activity D} = 15 - 12 = \mathbf{3 \text{ Weeks}}$$

- Activity D can experience up to **3 weeks of delay** without extending the overall project completion deadline of 15 weeks.
- **Critical Activity Sensitivities:**
 - Activities on the critical path (A, B, C, E, F, G) have **zero slack** ($\text{Slack} = 0$).
 - Any delay in any of these critical activities causes a direct, day-for-day delay in the overall 15-week delivery date.